Foundations in Climate & Energy Systems, UZH Carbon Footprint and Why it Matters

Decarb UZH: Experimental Approaches to Campus Decarbonization

Nils Handler

Universität Zürich

Spring Semester 2026

Global Climate Risks & Impacts

Key Climate Risks

- **Rising Temperatures:** Global average temperature has increased by 1.3-1.4°C since pre-industrial times
- **Sea Level Rise:** Global mean sea level has risen by 21cm since 1900
- **Extreme Weather:** Increased frequency and intensity of heatwaves, droughts, and heavy precipitation
- **Biodiversity Loss:** Climate change threatens 1 in 6 species globally

Switzerland particularly vulnerable

- **Climate Damages:** Global warming affects Switzerland twice as much as other regions, implying a cost of carbon is twice that of the global average (>400 CHF per ton)
- **Melting Glaciers:** The alps are a vulnerable ecosystem that is already changing as a result (cf. Blatten)
- **The Snow must Go on:** Skiing in the winter might soon be a distant memory

"Fighting Climate Change: International Attitudes toward Climate Policies." (AER, 2025)

Abstract

This paper explores global perceptions and understanding of climate change and policies, examining factors that influence support for climate action and the impact of different types of information. We conduct large-scale surveys with 40,000 respondents from 20 countries, providing new international data on attitudes toward climate change and respondents' socioeconomic backgrounds and lifestyles. We identify three key perceptions affecting policy support: perceived effectiveness of policies in reducing emissions, their impact on low-income households, and their effect on respondents' households (self-interest). Educational videos clarifying policy mechanisms increase support for climate policies; those merely highlighting climate change's impacts do not.

Citation

Dechezleprêtre, Antoine, Adrien Fabre, Tobias Kruse, Bluebery Planterose, Ana Sanchez Chico, and Stefanie Stantcheva. 2025. *American Economic Review* 115 (4): 1258–1300.

Science Paper: Knowing What Works

Why it's so hard to tell which climate policies actually work

Better tools are needed to analyse their effects

Science & technology | So much hot air

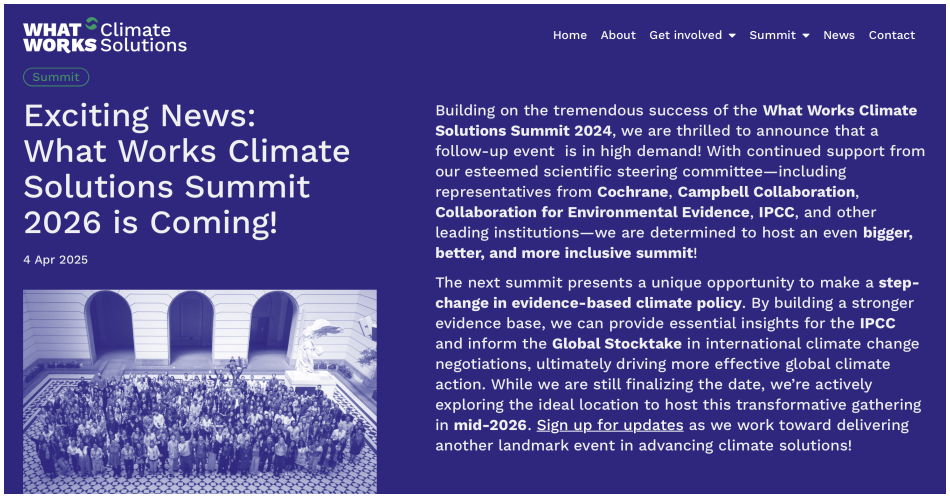
Why it's so hard to tell which climate policies actually work

Better tools are needed to analyse their effects

Share

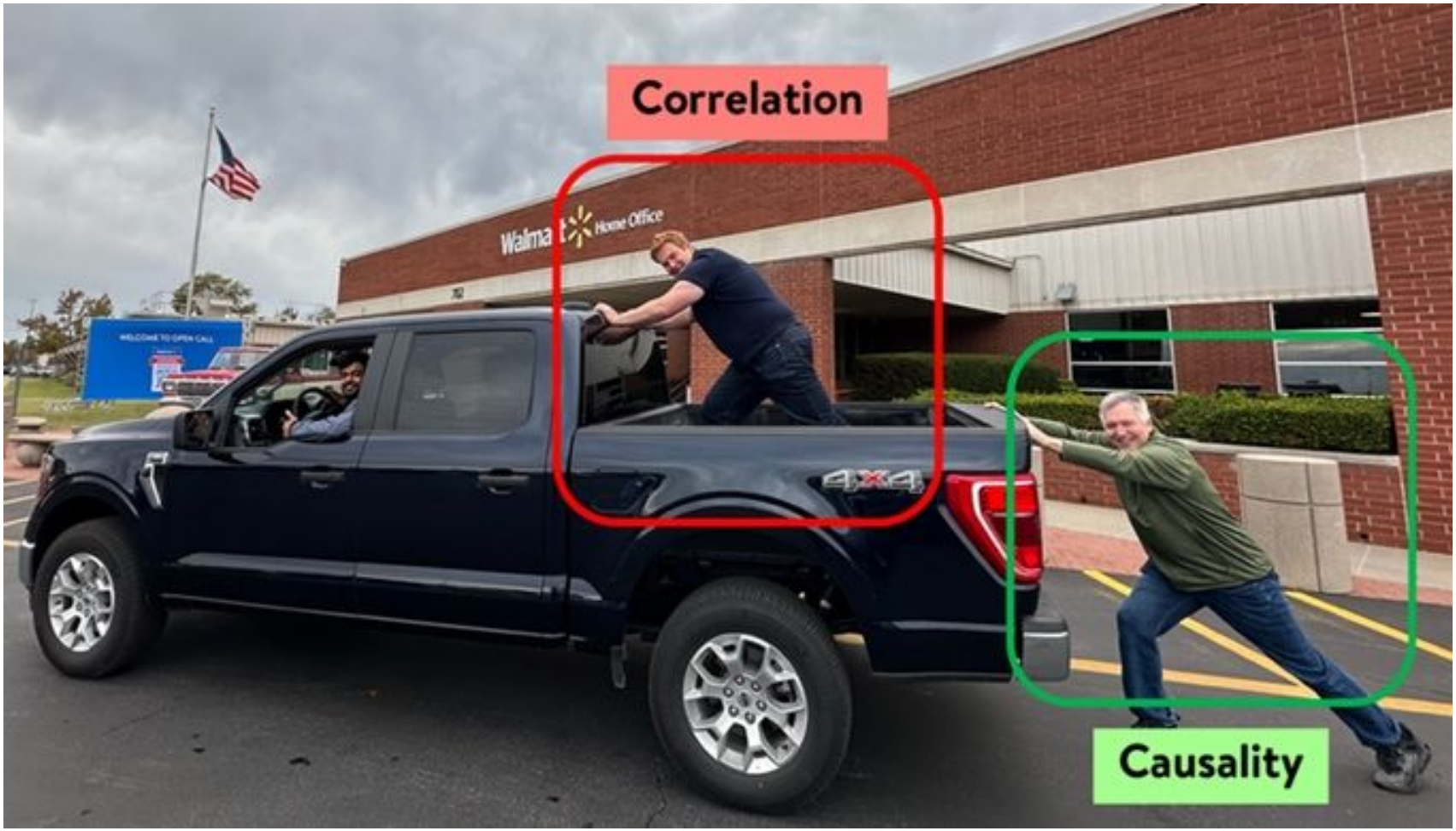


And systematically review the impact PHOTOGRAPH: GETTY IMAGES

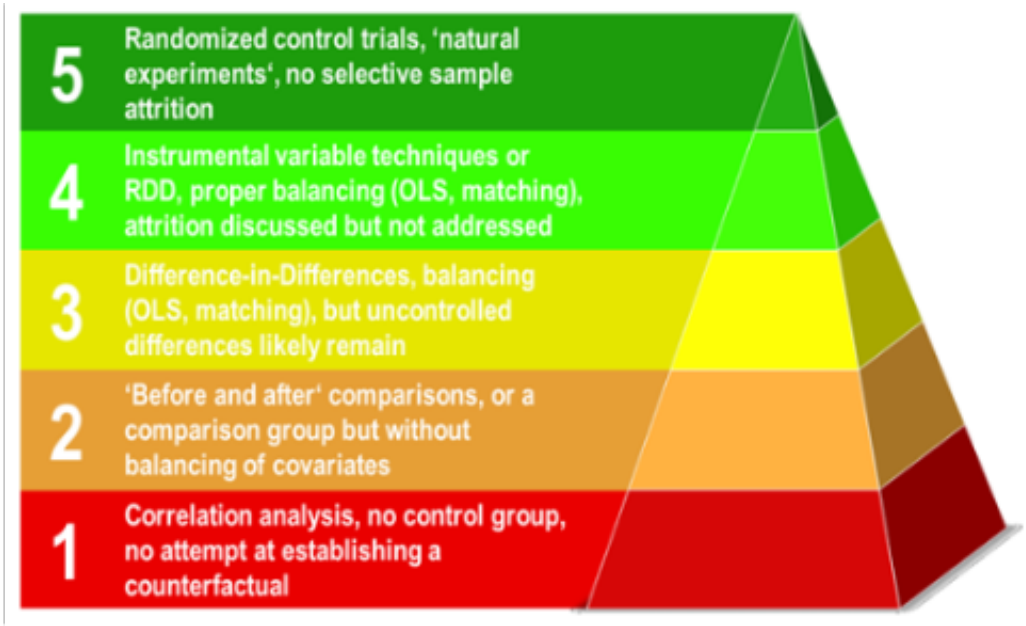


https://whatworksclimate.solutions/wwcs-summit-2026-is-coming/

This class in one slide



The value of RCTs from development to other fields of economics





What is Evidence-Based Policy-Making?

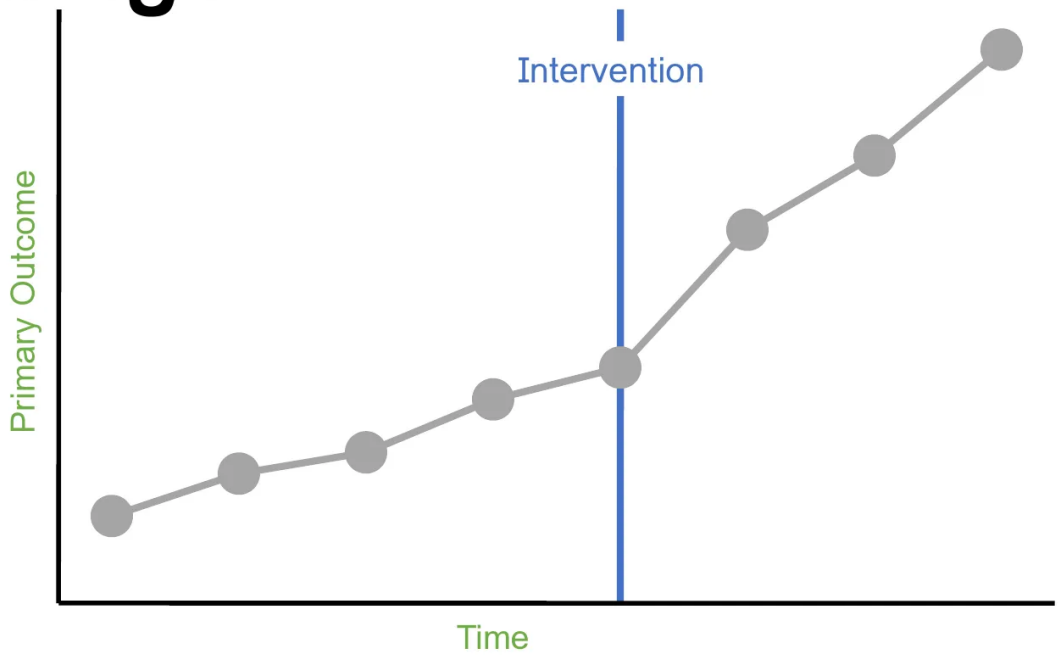
$$\text{Evidence} \rightarrow \text{Policy making} \rightarrow \text{Impact}$$

What typically happens:

- Policy-making focuses on matters other than impact, such as use of resources
- The evidence used for policy-making can not be used to support statements about effectiveness
- In public discourse, many causal claims are made based on non-causal evidence

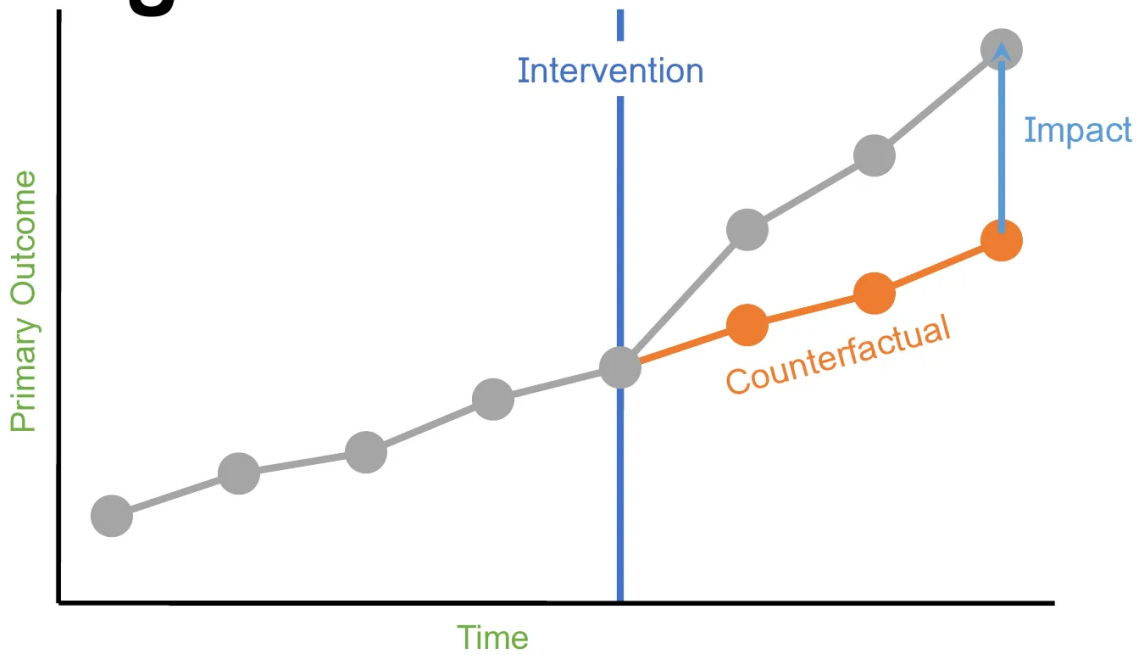
What do we mean by impact?

Was verstehen wir unter Wirkung?



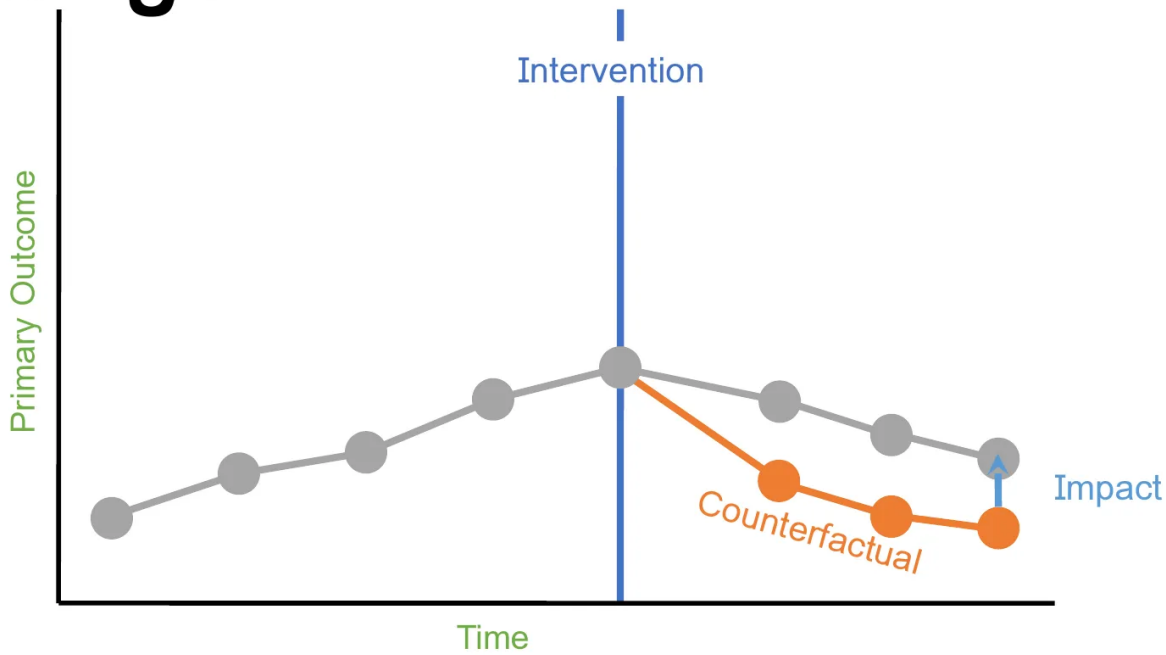
What do we mean by impact?

Was verstehen wir unter Wirkung?



What do we mean by impact?

Was verstehen wir unter Wirkung?

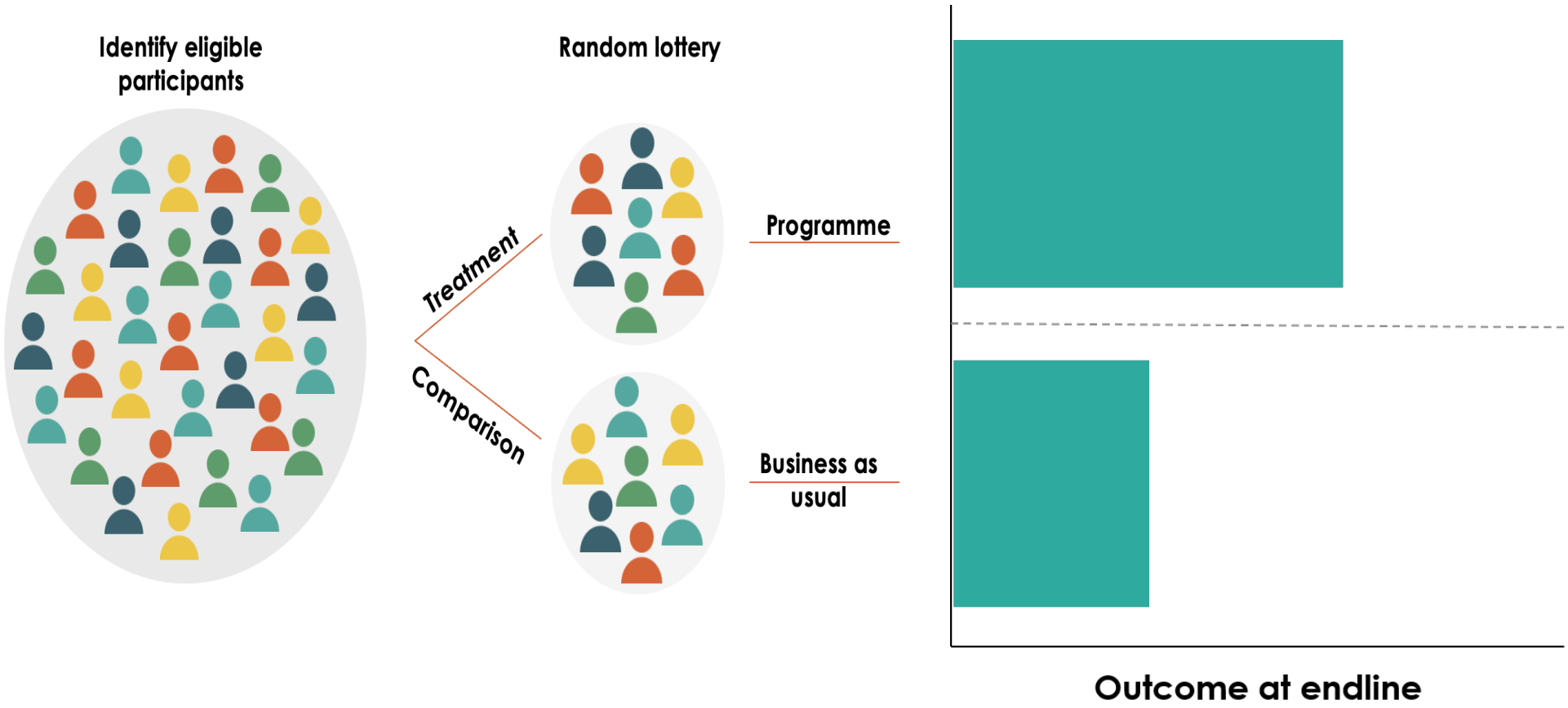


How can we measure impact?

Impact = counterfactual analysis
= Difference between the effect of a program and what occurs in the absence of that program

Problem: We can not directly measure what occurs in the counterfactual scenario, so we have to "reconstruct" it

Counterfactual construction through randomization

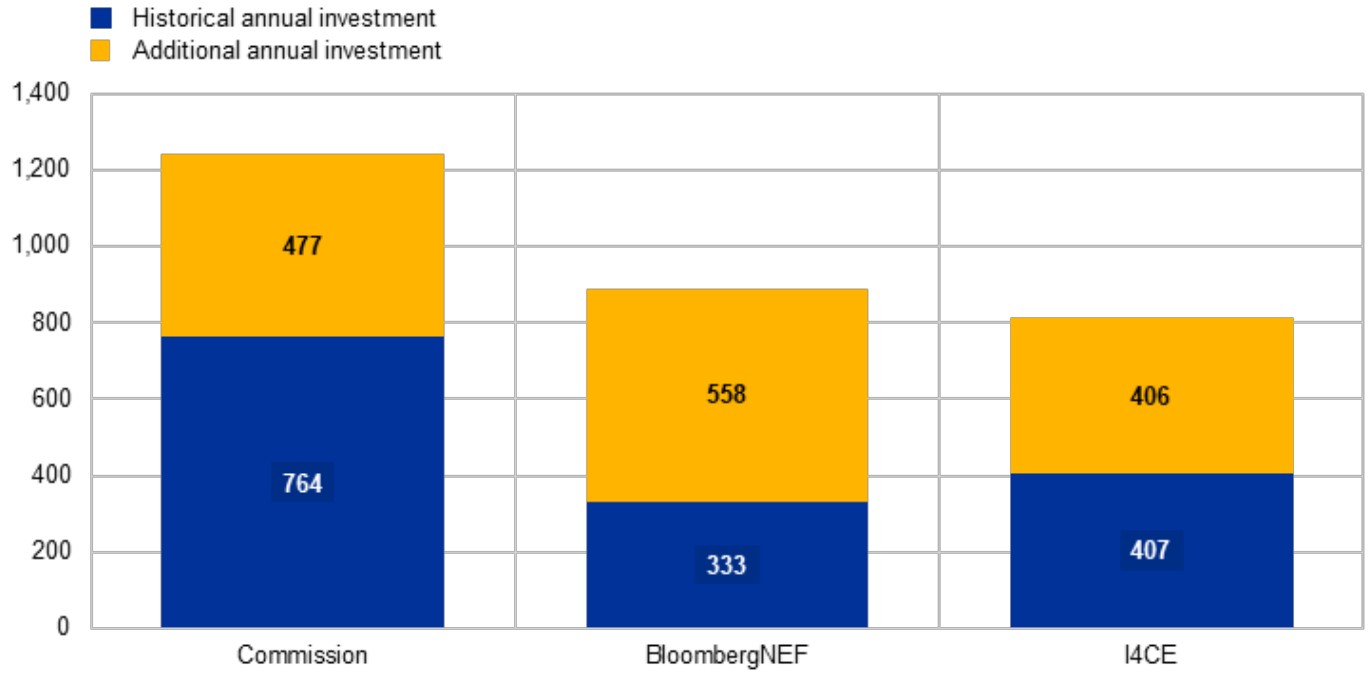


Key advantage of randomised evaluations:

Due to random assignment, members of the treatment and comparison groups do not differ systematically at the outset of the evaluation

Why is this relevant? Some Examples

- Nearly 50% more funding will be needed for Europe alone to achieve its net-zero targets (amounting to 8.3% of GDP annually), and this action will need to be based on evidence so as to not waste money.



- Current evidence can be flawed; Engineering models can over-predict the reductions that policies create, so assessing them afterwards is key.

Where else is this being applied?

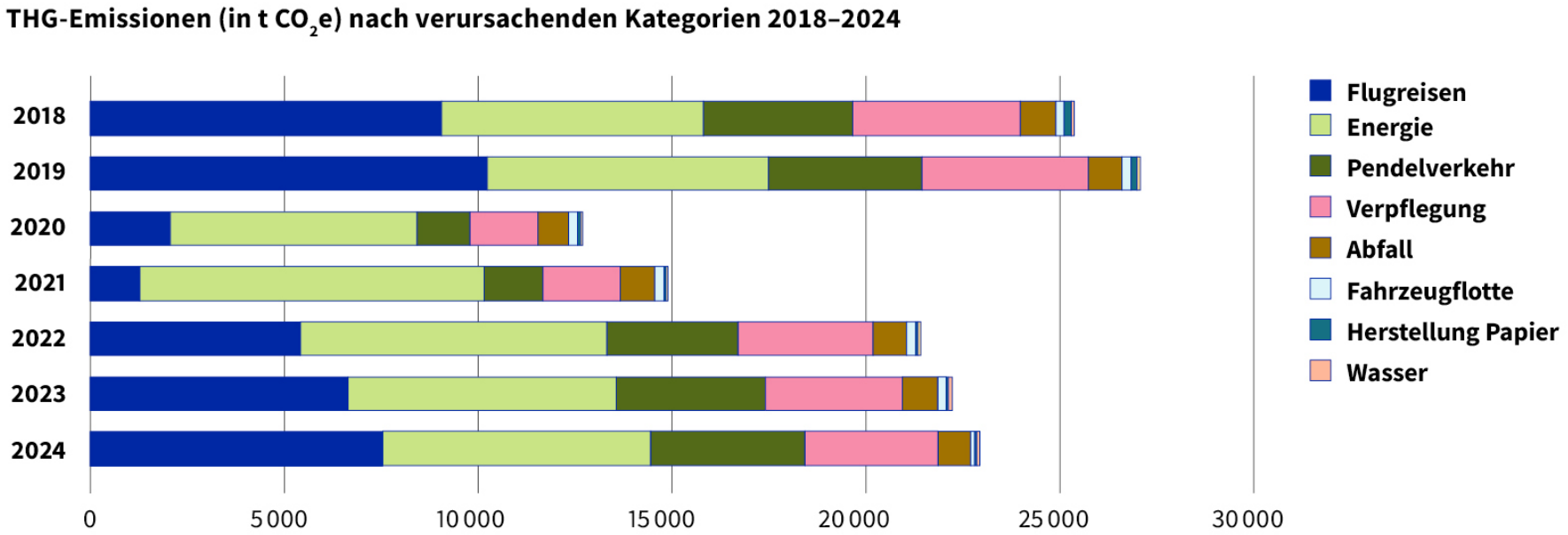
1. **J-PAL King Climate Action Initiative (K-CAI):**
   Mainly focused on developing countries

2. **Urban Labs Energy & Environment, University of Chicago**
   (Prof. Michael Greenstone)



UZH Energy Profile & Carbon Footprint

Carbon Footprint Breakdown



UZH at a Glance

- ~27,000 Students
- ~10,000 Staff
- ~80 Buildings

Brainstorming: What are potential areas for interventions?

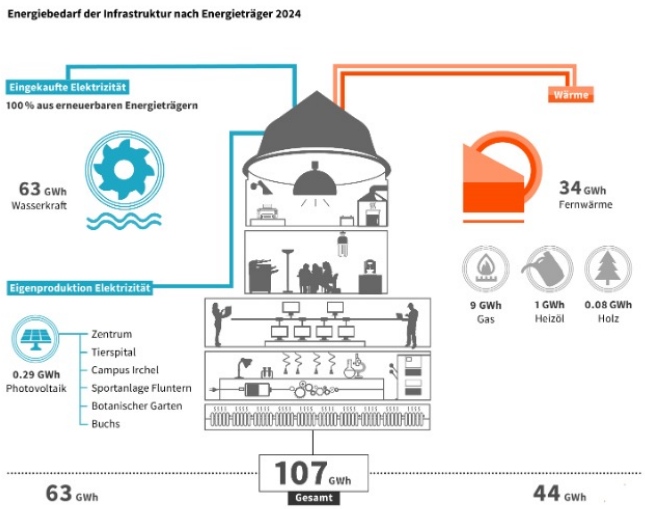
UZH Climate Goal

The University of Zurich has set itself the goal of becoming climate neutral by 2030. At least 50% of the reduction compared to 2018 is to be achieved through its own measures. The target covers the categories energy, air travel, commuting, vehicle fleet, catering, paper, water, and waste, and from 2035 additionally includes construction activities as well as laboratory and research materials.

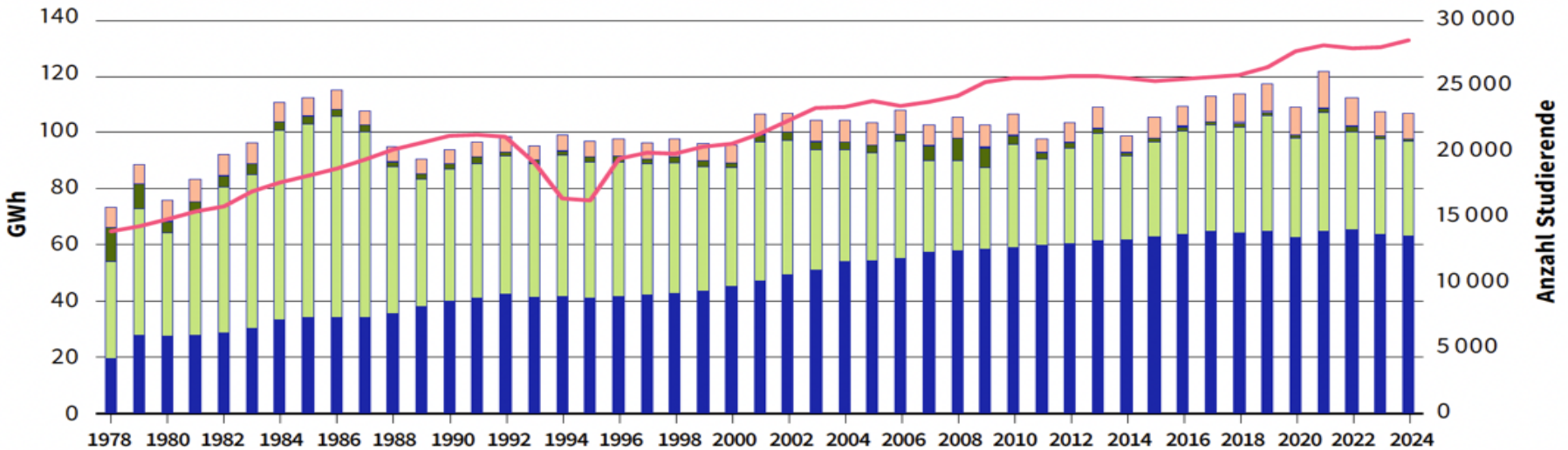
Building & Campus Energy Consumption

Campus Energy Characteristics

- Diverse Building Types
- High Occupancy Variability
- Research-Intensive Areas
- Aging Infrastructure
- Central Systems



Entwicklung des Energiebedarfs der UZH nach Energieträger (in GWh) 1978–2024



Building & Campus Energy Consumption

Diverse Building Types
Mix of offices, classrooms, laboratories, libraries, and residential buildings with different energy profiles

Key Consumption Drivers

- Building Envelope
- HVAC Systems
- Lighting
- Plug Loads
- Occupant Behavior

Next Intervention Opportunities

- **Technical:** Equipment upgrades, building retrofits, automation systems
- **Operational:** Optimized schedules, setpoint adjustments
- **Behavioral:** User education, feedback mechanisms
- **Structural:** Space utilization optimization

Potential source of inspiration; Prof. Robert D. Metcalf and MVPF

Robert D. Metcalfe

Professor of International and Public Affairs
SIPA Professorship of Energy Policy
Columbia University
robert.metcalfe@columbia.edu

Co-Editor: Journal of Public Economics
Associate Editor: Journal of Political Economy: Microeconomics
Associate Editor: Management Science
Research Associate: National Bureau of Economic Research (NBER)
Chief Economist: Centre for Net Zero, Octopus Energy Group
Faculty Affiliate, Center for Global Energy Policy
Faculty Affiliate: Abdul Latif Jameel Poverty Action Lab (J-PAL)

Curriculum Vitae | Google Scholar | RePEc | Twitter | LinkedIn | Bluesky



Fields: Applied microeconomics; environmental & energy economics; behavioral & experimental economics; labor/personnel economics; public economics.

Selected Peer-Reviewed Publications:

Do the Effects of Social Nudges Persist? Theory and Evidence from 38 Natural Field Experiments in Energy Conservation
Brandon, A., Ferraro, P., List, J., Metcalfe, R., Price, M., Rundhammer, F. Forthcoming, ***Review of Economic Studies***

Nudging Timely Wage Reporting: Field Experimental Evidence from the U.S. SSI Program
Zhang, C.Y., Hemmeter, J., Kessler, J.B., Metcalfe, R., Weathers, R. (2023) ***Management Science***
[Ungated PDF]

The Making of Homo Honoratus: From Omission to Commission
Hallsworth, M., List, J., Metcalfe, R., Vlaev, I. (2023) ***Journal of Consumer Psychology***
[Ungated PDF]

Measuring the Welfare Effects of Shame and Pride
Butera, L., Morrison, W., Metcalfe, R., Taubinsky, D. (2022) ***American Economic Review***
[Ungated PDF] [Slides] [Click for Charity user interface] [Download QSF files for Prolific version]

Fatalism, Beliefs, and Behaviors During the COVID-19 Pandemic
Akesson, J., Ashworth-Hayes, S., Hahn, R., Metcalfe R., Rasooly, I. (2022) ***Journal of Risk and Uncertainty***

Efficiency and Equity Impacts of Energy Subsidies.
Hahn, R., Metcalfe, R. (2021) ***American Economic Review***
[Ungated PDF] [Slides]

Understanding the Response to Financial and Non-financial Incentives in Education: Field Experimental Evidence using High-stakes Assessments
Burgess, S., Metcalfe, R., Sadoff, S. (2021) ***Economics of Education Review***
[Ungated PDF]

The Impact of Management Practices on Employee Productivity: A Field Experiment with Airline Captains.
Gosnell. G., List, J., Metcalfe, R. (2020) ***Journal of Political Economy*** Lead article
[Harvard Business Review] [Ungated PDF] [Slides]

Working Papers:

Expecting Climate Change: Evidence from a Nationwide Field Experiment in the Housing Market
Fairweather, D., Kahn, M., Metcalfe, R., Sandoval-Olascoaga, S. *Revise and resubmit*, ***American Economic Review***.

A Welfare Analysis of Policies Impacting Climate Change
Hahn, R., Hendren, N., Metcalfe, R., Sprung-Keyser, B. *Revise and resubmit*, ***American Economic Review***.

Management and Productivity in Retail
Metcalfe, R., Sollaci, A., Syverson, C. *Revise and resubmit*, ***Journal of Political Economy***

The Value of Time in the United States: Estimates from Nationwide Natural Field Experiments
Goldszmidt, A., List, J., Metcalfe, R., Muir, I., Smith, K., Wang, J. *Revise and resubmit*, ***American Economic Journal: Economic Policy***

The Human Perils of Scaling Smart Technologies: Evidence from Field Experiments
Brandon, A., Clapp, C., List, J., Metcalfe, R., Price, M. *Revise and resubmit*, ***American Economic Journal: Economic Policy***

Decarbonizing Heat: The Impact of Heat Pumps and a Time-of-Use Heat Pump Tariff on Energy Demand
Bernard, L., Hackett, A., Metcalfe, R., Schein, A. [Executive Summary] *Revise and resubmit*, ***American Economic Journal: Economic Policy***

Do Water Audits Work?
Akesson, J., Hahn, R., Kochhar, R., Metcalfe, R. *Revise and resubmit*, ***Economic Journal***.

Race and Redistribution in the United States: An Experimental Analysis
Akesson, J., Hahn, R., Metcalfe, R., Rasooly, I. *Revise and resubmit*, ***Journal of Public Economics***

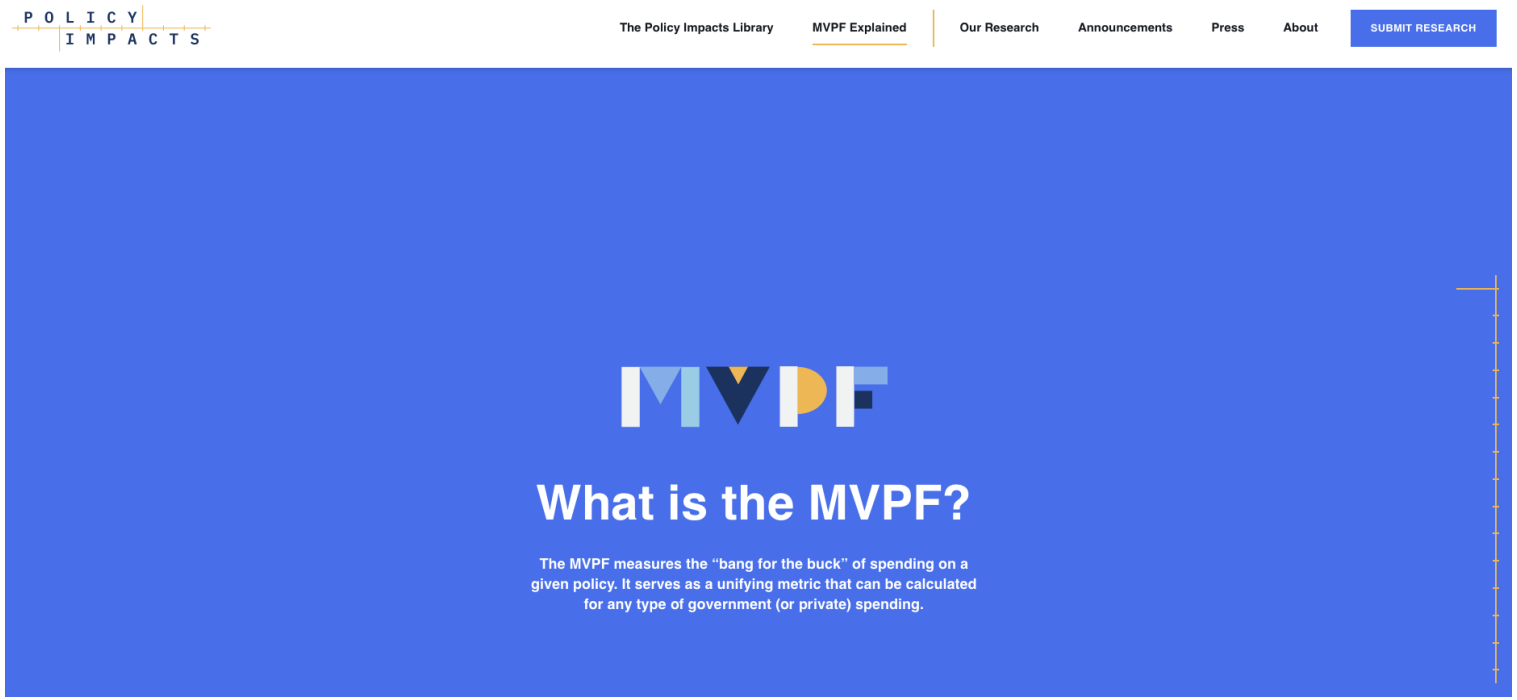
Smart Meters: Do Their Prices Matter to Their Adoption and Do They Save Energy?
List, J., Metcalfe, R., Price, M. *Revise and resubmit*, ***Journal of Environmental Economics and Management***

Can Myopic Loss Aversion Explain the Equity Premium Puzzle? Evidence from a Natural Field Experiment with Professional Traders
Larson, F., List, J., Metcalfe, R. *Reject and resubmit*, ***Econometrica***

The Impact of Dynamic Prices on Electric Vehicle Public Charging Demand: Evidence from a Nationwide Natural Field Experiment

Potential source of inspiration; Prof. Robert D. Metcalf and MVPF

MVPF: Marginal Value of Public Funds, a measure of the impact of additional public spending on an initiative



https://policyimpacts.org/mvpf-explained/what-is-the-mvpf/

Course Projects: RCT Development

RCT Development

Design an experiment that will test an intervention relevant to decarbonization.

Learning Objectives: Students will learn about the challenges related to theory construction, hypothesis and intervention design, and logistical constraint consideration

Submission: A package describing the experimental design consisting of a pre-analysis plan, an ethics approval form, and a pre-registration

Course Projects: Literature Review

Literature Review

Review the literature of a given (sub-)field connected to decarbonization and identify knowledge gaps.

Learning Objectives: Students will develop meta-analysis skills to summarize, measure, and critique the understood effect of current interventions and the potential gaps or issues with the research behind those results.

Submission: A literature review with a well-defined scope that clearly outlines the current state of the literature and issues/gaps in that literature. Relevant pre-registration documents will be decided upon project decision.

Now its time for a. . . .



As inspired by NESTA

Week 2 Preview: RCTs

Coming Next Week

- Randomized Controlled Trials & Policy Evaluation
- Causal Inference
- Randomization
- Treatment Effects

Preparation for Next Week

- **Required Reading:** Pless et al. Bringing Rigor to Energy Innovation Policy, Nature energy
- **Case Study:** Fowlie et al. Do Energy Efficiency Investments Deliver? QJE, 2018
- **Reflection Question:** How might randomized controlled trials be applied to campus energy interventions?